

# COMP3023 Design and Analysis of Algorithms

Spring 2026

## Assignment 1

### Question 1:

Given the input  $A = [2, -7, 3, 1, 2, -3, 5, -4, 6, 2]$ , track the divide-and-conquer algorithm to find the maximum contiguous subarray. Construct the recursive calls as a tree. (20pt)

### Question 2:

Given two polynomials  $A(x) = 2 + 3x + x^2 + 2x^3 + x^4$  and  $B(x) = x + 4x^2 + 2x^4$ , track the  $\mathcal{O}(n^{\log 3})$  algorithm to calculate  $A(x) \times B(x)$ . Construct the recursive calls as a tree. (25pt)

### Question 3.

Please use the method in *DSelection* to find the median of medians recursively on the following array  $A$  in groups of 5. To unify marking, the median of  $n$  elements is the  $\lceil \frac{n}{2} \rceil$ -th smallest element. (15pt)

$A =$  (22,33,17,23,9,31,16,21,26,4,1,13,3,24,20,14,2,30,18,6,40,15,5,38,35,7,  
12,29,34,27,36,39,8,28,32,19,37,25,10,11)

#### Question 4.

Present the deterministic linear-time selection algorithm  $DSelection(A, p, r, i)$  in Lecture 5 as pseudo codes. The interface of  $DSelection$  is as

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**Algorithm 1.**  $DSelection(A, p, r, i)$ 

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**Input:**  $A$ , an array of integers;  
           $p$ , the minimum index of  $A$ ;  
           $r$ , the maximum index of  $A$ ;  
           $i$ , an integer.  
**Output:** the  $i$ -th smallest element in  $A$ .

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You are allowed to make function calls to  $Partition$ , which is defined as

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**Algorithm 2.**  $Partition(A, p, r, m)$ 

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**Input:**  $A$ , an array of integers;  
           $p$ , the minimum index of  $A$ ;  
           $r$ , the maximum index of  $A$ ;  
           $m$ , the index of the specified pivot.  
**Output:** an integer  $x$  of the permutation  $A'$  of  $A$ , such that the input array  $A$  becomes  $A'$  after algorithm execution,  $A'[i] \leq A'[x]$  for all  $p \leq i < x$  and  $A'[j] > A'[x]$  for all  $x < j \leq r$ .

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Please also pay attention to the interface of Algorithm 2.

- The input needs one extra argument  $m$ , to specify which element is selected as the pivot (not randomly, not the last element).
- The output is the index of the pivot (after permutation) instead of its value.
- After executing the algorithm, the input array  $A$  becomes  $A'$  (as a side-effect), which satisfies the partition property.

To simplify your pseudocodes, you are also to use  $GroupF$  and  $Median$ , which are defined as

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**Algorithm 3.**  $GroupF(A, p, r)$ 

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**Input:**  $A$ , an array of integers;  
           $p$ , the minimum index of  $A$ ;  
           $r$ , the maximum index of  $A$ .  
**Output:** an array  $G = [g_1, g_2, \dots, g_{\lfloor \frac{n}{5} \rfloor}]$  such that each  $g_i = A[p + 5(i - 1), \dots, p + 5i]$  and  $g_{\lfloor \frac{n}{5} \rfloor}$  contains the last few elements of  $A$ .

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**Algorithm 4.**  $Median(A)$ 

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**Input:**  $A$ , an array of integers.  
**Output:** the median  $m$  of  $A$ .

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These specifications will make your algorithm consistent. (20pt)

### Question 5.

Mr. Trivial wants to design an algorithm to find MCS. He has realized the following.

#### Observation 1.

Each array  $A[0 \dots n - 1]$  can be split into three subarrays  $L = A[0 \dots i - 1]$ ,  $M = A[i \dots j]$ , and  $R = A[j + 1 \dots n - 1]$  such that  $0 \leq i \leq j \leq n - 1$  (when  $i = 0$ ,  $L$  is an empty array; and same for  $j = n - 1$ ).

#### Observation 2.

If  $M$  is the MCS of  $A$ , then  $L$  is the minimum prefix of  $A$  and  $R$  is the minimum suffix of  $A$ .

#### Example.

For example,  $A = [1, -3, 4, 2, -2, 1]$ . The minimum prefix is  $L = [1, -3]$ ; the minimum suffix is  $R = [-2, 1]$ ; and the MCS is  $M = [4, 2]$ .

Then, Mr. Trivial's algorithm is simply as (in natural language)

1. Find the minimum prefix of  $A$ , called  $L$ .
2. Find the minimum postfix of  $A$ , called  $R$ .
3. Remove  $L$  and  $R$  from  $A$ . The remained subarray is the MCS of  $A$ .

However, these two observations are not generally true. The algorithm is of course incorrect. Thus,

- a) Prove that Observation 2 is false. (5pt)
- b) Claim a correct observation. (5pt)
- c) Use the observation in Part b) to design a divide-and-conquer algorithm to find the MCS. Your algorithm should
  - run in  $\mathcal{O}(n \log n)$  time;
  - **must** use *MinPrefix()* and *MinSuffix()* which find the value of the minimum prefix and suffix of an input array in  $\mathcal{O}(n)$  time (assuming they are already implemented);
  - **cannot** use or implement *MaxPrefix()* and *MaxSuffix()* (Algorithm 3 on page 16 of Lecture 02); and
  - be presented as pseudo codes. (5pt)
- d) Show your algorithm runs in  $\mathcal{O}(n \log n)$  time. (5pt)